

SULIT



SECP2523 DATABASE

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ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	Koperasi Kakitangan KADA (KKK)
System Name	KKK Online System
Group Name	Tech Hi-Five

Table of Contents

1.1 Overall Description	2
1.2 Project Weaknesses or Improvement	3
1.3 Proposed Module	3
2.1 Conceptual ERD	3
2.2 Logical ERD	4
2.3 Data Dictionary for Logical ERD	5
2.3.1 Identifying Entity	5
2.3.2 Identifying Relationship Type	7
2.3.3 Attributes Description	7
2.4 Relational Database Schema	12
2.5 Normalization	13

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

KKK Online System is created for the KADA company as the user management system for several use case including but not limited to member application, loan application, administration and financial reporting. Digitizing the processes will improve functionality, efficiency and reduced time consumed compared to traditional paper-based system. It also offers secure and authentic functions for both the admin and user and records are kept in a secure and easy to manage way.

The system is created for new users, members and admin of KKK system. **New user** represents the employees of *Koperasi Kakitangan KADA Kelantan Berhad* that have not register themselves as member that uses the system to register for membership. **Members** of KKK system enjoys the benefits and has access to the member functions of the system. **Admins** are the KKK employee that manage and control the content, policy and status of the member.

This system currently contains 6 modules. The non administrative module includes **User module** which includes apply for membership, log in , **Member module** contains update member profile, view financial status and transaction records, **Loan module** that is apply loan and view loan application status. The administrative module includes **Admin Approval Module** for viewing list of applicants, approve membership application and loan application and **Admin update module** for administer which includes update banner, update policies and record transactions.

The module distribution of the project are shown as below:

Name	Matric Number	Module
Tan Yi Ya	A23CS0187	User Module
Chua Jia Lin	A23CS0069	Member Module
Teh Ru Qian	A23CS0191	Loan Module, Admin Reporting Module
Goe Jie Ying	A23CS0224	Admin Approval Module
Lam Yoke Yu	A23CS0233	Admin Updating Module

1.2 Project Weaknesses or Improvement

Upon careful consideration of the system, there is some lack of our system that needs

to be improved. There is a weakness of our proposed KKK Online System, that is missing the functionality of member accessing frequently asked questions to view answers, and admin providing supports to the new member that are unfamiliar with the system. Whether it is about how to use the system or what are the rules of applying for certain loans, the members did not have any system-wise choice to ask about anything.

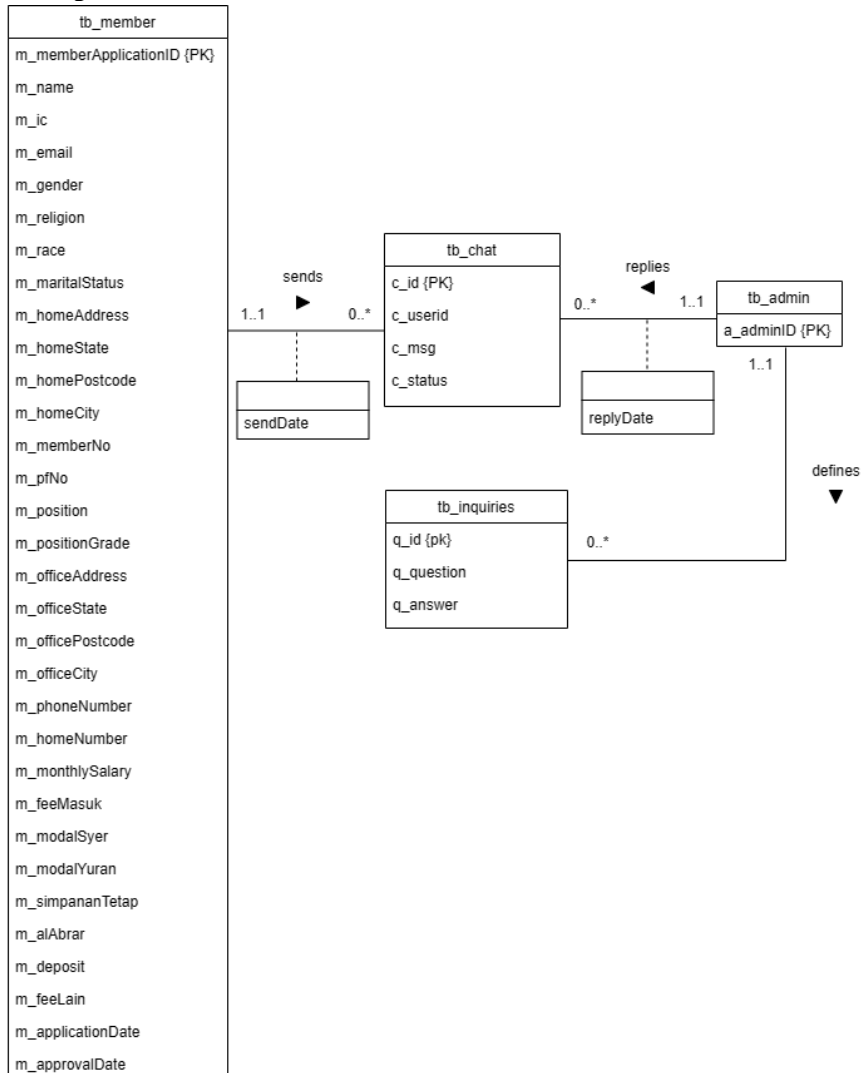
Improvement that can be made are adding functionality for members to access a chat window in all of the page, initially shown as a small icon in the bottom left corner of the screen. Upon clicking the icon, the users are shown with some predefined questions that are frequently asked by user. When clicking the questions they are replied with the predefined answers to aid them of their unfamiliarity. If the problem or questions are not listed among the predefined questions, user can typed in the chat box manually and an online admin will reply to their questions.

1.3 Proposed Module

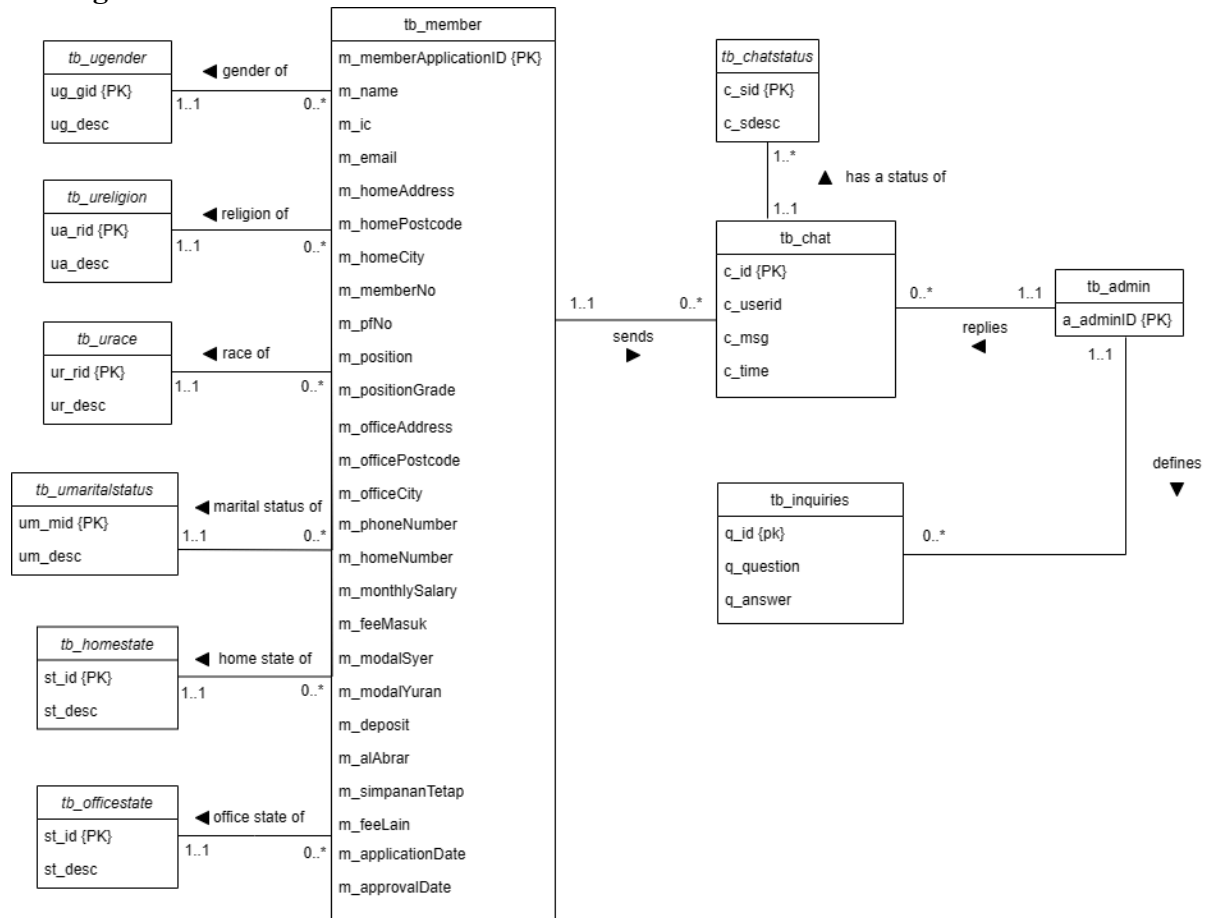
The new module proposed is live chat module. Live chat module consist of use case for member such as view frequently asked questions and send inquiries to admin. For admin, they can view user questions and reply to user questions. This module is designed to overcome the problem of lack of communication between member of KKK system and admin.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

2.3.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
tb_member	General term describing all user logged in as member using KKK Online System	KKK Member, Member Application	Each member has one gender, one religion, one race and one marital status only. Each member resides and works in one state. Each member can become guarantors for other members. Each member applies for many loans. Each member application will have one status only at

			any one time. Each member owns one financial record. Each member application is approved by one admin. Each member can make many transactions. Members are one kind of user for the system. Each member can assign many heirs.
tb_ugender	General term describing the gender of a member	Gender	Each gender can be shared by many members.
tb_ureligion	General term describing the religion of a member	Religion	Each religion can be shared by many members.
tb_urace	General term describing the race of a member	Race	Each race can be shared by many members.
tb_umaritalstatus	General term describing the marital status of a member	Marital status	Each marital status can be shared by many members.
tb_homestate	General term describing the home state of a member	State	There can be none or many members residing in the state.
tb_officestate	General term describing the office state of a member	State	There can be none or many members working in the state.
tb_admin	General term describing all user logged in as admin using KKK Online System	KKK Staff	Admin is one type of user. Admin can edit many banners. Each admin can generate many reports. Each admin alters many policies. Each admin controls many loan applications. Each admin approves

			many member applications. Each admin can record many transaction records.
tb_chatstatus	General term describing status of chat message	-	Each chat message can have a single status at one time.
tb_chat	General term describing all messages	Chat messages	Each admin and member can send one or more messages. While each message can only be sent by 1 member or admin.
tb_inquiries	General term describing all predefined questions	-	Each predefined questions are defined by a single admin only.

2.3.2 Identifying Relationship Type

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
tb_member	1..1 0..* 0..* 0..* 0..* 0..* 0..*	sends links is religion of is race of has a marital status of resides works in	tb_chat tb_ugender tb_ureligion tb_urace tb_umaritalstatus tb_homestate tb_officestate	0..* 1..1 1..1 1..1 1..1 1..1 1..1
tb_admin	1..1 1..1	replies defines	tb_chat tb_inquiries	0..* 0..*
tb_chat	1..1	has a status of	tb_chatstatus	1..*

2.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
tb_member	memberApplication ID (PK)	int (11)	Uniquely identifies each member	No	No	1001

			application			
	m_name	varchar(50)	User's full name	No	No	Ali bin Abu
	m_ic	varchar(14)	User's identification number	No	No	850101-08-1033
	m_home Address	varchar(255)	User's home address	No	No	1, Taman Bahagia, Jalan Bahagia,
	m_homePostcode	int (11)	User's home postcode	No	No	57000
	m_home City	varchar (50)	User's home city	No	No	Kuala Lumpur
	m_memberNo	int	Uniquely identifies each member	No	No	1
	m_pfNo	int	User's PF number	No	No	1001
	m_position	varchar(50)	User's position	No	No	Pengurus
	m_positionGrade	varchar(11)	User's position grade	No	No	G41
	m_office Address	varchar(255)	User's office address	No	No	25, Jalan Makmur
	m_office Postcode	int (11)	User's office postcode	No	No	57000
	m_office City	varchar (50)	User's office city	No	No	Kuala Lumpur
	m_phone Number	varchar(13)	User's phone number	No	No	0123456789
	m_home	varchar(13)	User's	Yes	No	051234567

	Number	)	home number			
	m_monthlySalary	double	User's monthly salary	No	No	3500.00
	m_feeMasuk	double	User's entry fee	No	No	45.00
	m_modalSyer	double	User's shares	No	No	300.00
	m_modalYuran	double	User's share fee	No	No	50.00
	m_deposit	double	User's deposit	No	No	50.00
	m_alAbrar	double	A kind of user's fund	No	No	5.00
	m_simpanTetap	double	User's fixed saving	No	No	0.00
	m_feeLain	doube	User's other fee	No	No	0.00
	m_applicationDate	datetime	The date when user submit his/her member application	No	No	2025-01-03 16:00:00
	m_approvalDate	timestamp	The date when the member application being approved by admin	Yes	No	2025-01-15 06:40:00
tb_ugender	ug_gid (PK)	int (11)	The unique id for gender	No	No	1
	ug_desc	varchar (10)	User's gender description	No	No	Lelaki

tb_ureligion	ua_rid (PK)	int (11)	The unique id for religion	No	No	1
	ua_desc	varchar (10)	User's religion description	No	No	Islam
tb_urace	ur_rid (PK)	int (11)	The unique id for race	No	No	1
	ur_desc	varchar (10)	User gender's description	No	No	Lelaki
tb_umarital status	um_mid (PK)	int (11)	The unique id for marital status	No	No	1
	um_desc	varchar (10)	User's marital status description	No	No	Bujang
tb_homestat e	st_id (PK)	int (11)	The unique id for user's home state	No	No	1
	st_desc	varchar (15)	User's home state description	No	No	Johor
tb_officesta te	st_id (PK)	int (11)	The unique id for user's office state	No	No	1
	st_desc	varchar (15)	User's office state description	No	No	Melaka
tb_admin	a_adminI D (PK)	int (11)	Uniquely identifies each admin	No	No	200
tb_chat	c_id(PK)	int(11)	The unique id for chat messages	No	No	1

	c_userid	int(11)	The user id of the user sending the message	No	No	1000
	c_msg	vachar(1000)	The actual message that user sends	No	No	Apakah syarat minimum untuk memohon pinjaman?
	c_time	timestamp	The time of user send message	No	No	2025-01-15 06:40:00
tb_inquiries	q_id (PK)	int(11)	The unique id for predefined inquiries	No	No	1
	q_question	vachar(1000)	The predefined question	No	No	Bagaimana kah cara untuk mengemas kini maklumat saya?
	q_answer	vachar(1000)	The answer to the question	No	No	Tekan pada profil, kemudian pilih kemas kini maklumat dan buat perubahan yang diperlukan

						. Apabila selesai, tekan simpan.
tb_chatstatus	c_sid (PK)	int(11)	The unique id for the status of chat message	No	No	1
	c_sdescs	vachar(15)	The chat message status description	No	No	Dihantar

2.4 Relational Database Schema

tb_admin (a_adminID)

PK: a_adminID

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID

FK: m_gender references tb_ugender (ug_gid)

FK: m_religion references tb_ureligion (ua_rid)

FK: m_race references tb_urace (ur_rid)

FK: m_maritalStatus references tb_umaritalstatus (um_mid)

FK: m_status references tb_status (s_sid)

FK: m_homeState references tb_homestate (st_id)

FK: m_officeState references tb_officestate (st_id)

FK: m_adminID references tb_admin (a_adminID)

tb_homestate (st_id, st_desc)

PK: st_id

tb_officestate (st_id, st_desc)

PK: st_id

tb_ugender (ug_gid, ug_desc)

PK: ug_gid

tb_umaritalstatus (um_mid, um_desc)

PK: um_mid

tb_urace (ur_rid, ur_desc)
PK: ur_rid

tb_ureligion (ua_rid, ua_desc)
PK: ua_rid

tb_chat (c_id, c_userid, c_msg, c_time, c_status)
PK: c_id
FK: c_status references tb_chatstatus(c_sid)

tb_chatstatus (c_sid, c_sdesc)
PK: c_sid

tb_inquiries(q_id, q_question, q_answer)
PK: q_id

2.5 Normalization

1NF

tb_admin (a_adminID)
PK: a_adminID

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (a_adminID)

tb_chat (c_id, c_userid, c_msg, c_time, c_status)
PK: c_id

tb_chatstatus (c_sid, c_sdesc)
PK: c_sid

tb_inquiries(q_id, q_question, q_answer)
PK: q_id

2NF

tb_admin (a_adminID)
PK: a_adminID

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState,

m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID

FK: m_gender references tb_ugender (ug_gid)

FK: m_religion references tb_ureligion (ua_rid)

FK: m_race references tb_urace (ur_rid)

FK: m_maritalStatus references tb_umaritalstatus (um_mid)

FK: m_status references tb_status (s_sid)

FK: m_homeState references tb_homestate (st_id)

FK: m_officeState references tb_officestatus (st_id)

FK: m_adminID references tb_admin (a_adminID)

tb_homestate (st_id, st_desc)

PK: st_id

tb_offisestate (st_id, st_desc)

PK: st_id

tb_ugender (ug_gid, ug_desc)

PK: ug_gid

tb_umaritalstatus (um_mid, um_desc)

PK: um_mid

tb_urace (ur_rid, ur_desc)

PK: ur_rid

tb_ureligion (ua_rid, ua_desc)

PK: ua_rid

tb_chat (c_id, c_userid, c_msg, c_time, c_status)

PK: c_id

FK: c_status references tb_chatstatus(c_sid)

tb_chatstatus (c_sid, c_sdesc)

PK: c_sid

tb_inquiries(q_id, q_question, q_answer)

PK: q_id

3NF

No transitive dependency present as all non-key attributes directly depend on their respective primary key. None of the attributes depend on other non-key attributes, hence satisfying 3NF.

tb_admin (a_adminID)

PK: a_adminID

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID

FK: m_gender references tb_ugender (ug_gid)

FK: m_religion references tb_ureligion (ua_rid)

FK: m_race references tb_urace (ur_rid)

FK: m_maritalStatus references tb_umaritalstatus (um_mid)

FK: m_status references tb_status (s_sid)

FK: m_homeState references tb_homestate (st_id)

FK: m_officeState references tb_officestate (st_id)

FK: m_adminID references tb_admin (a_adminID)

tb_homestate (st_id, st_desc)

PK: st_id

tb_officestate (st_id, st_desc)

PK: st_id

tb_ugender (ug_gid, ug_desc)

PK: ug_gid

tb_umaritalstatus (um_mid, um_desc)

PK: um_mid

tb_urace (ur_rid, ur_desc)

PK: ur_rid

tb_ureligion (ua_rid, ua_desc)

PK: ua_rid

tb_chat (c_id, c_userid, c_msg, c_time, c_status)

PK: c_id

FK: c_status references tb_chatstatus(c_sid)

tb_chatstatus (c_sid, c_sdesc)

PK: c_sid

tb_inquiries(q_id, q_question, q_answer)

PK: q_id